Week 7

Consider vectors $u = \begin{bmatrix} 1 & -2 & -1 \end{bmatrix}$ and $v = \begin{bmatrix} -2 & 1 & -1 \end{bmatrix}$

1. Compute $||u||$ and $||v||$

$$||u|| = \sqrt{u \cdot u} = \sqrt{1 \cdot 1 + (-2)(-2) + (-1)(-1)} = \sqrt{6}$$

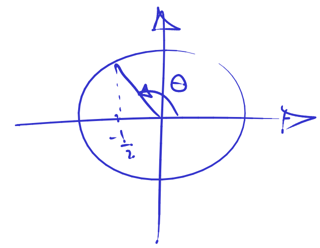
$$||v|| = \sqrt{v \cdot v} = \sqrt{(-2)(-2) + 1 \cdot 1 + (-1)(-1)} = \sqrt{6}$$

2. Compute the dot product between u and v

$$u \cdot v = v \cdot u = 1\,(-2) + (-2)(1) + (-1)(-1) = -3$$

3. Compute angle between vectors u and v

$$\cos\theta = \frac{u \cdot v}{\|u\|\|v\|} = \frac{-3}{\sqrt{6}\sqrt{6}} = \frac{-1}{2}$$



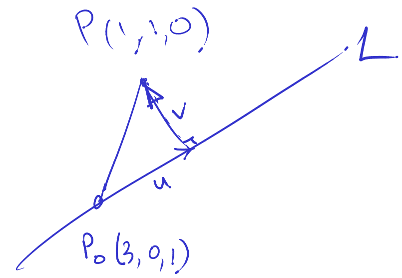
$$\Rightarrow \theta = \frac{2\pi}{3}$$

4. Find the projection of v onto u

$$\text{Proj}_u v = \frac{u \cdot v}{u \cdot u} u = \frac{-3}{6} [1 \quad -2 \quad -1]$$

$$= \left[\frac{-1}{2} \quad 1 \quad \frac{1}{2}\right]$$

5. Find the shortest distance between $P = (1, 1, 0)$ and the line
$L = \begin{bmatrix} 3 & 0 & 1 \end{bmatrix} + t \underbrace{\begin{bmatrix} 1 & -2 & -1 \end{bmatrix}}_{d}$



an arbitrary point on L

$$\text{dist}(P, L) = \|v\|$$

$$v = \overrightarrow{P_oP} - u$$

$$u = \text{Proj}_d \, P_oP$$

$$P_oP = \begin{bmatrix} -2 & 1 & -1 \end{bmatrix}$$

$$\Rightarrow u = \frac{P_oP \cdot d}{d \cdot d} d = \frac{-2-2+1}{1+4+1} \begin{bmatrix} 1 & -2 & -1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & 1 & \frac{1}{2} \end{bmatrix}$$

$$\Rightarrow v = \begin{bmatrix} -2 & 1 & -1 \end{bmatrix} - \begin{bmatrix} -\frac{1}{2} & 1 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} -\frac{3}{2} & 0 & -\frac{3}{2} \end{bmatrix}$$

$$\Rightarrow \|v\| = \sqrt{v \cdot v} = \sqrt{\frac{9}{4} + 0 + \frac{9}{4}} = \sqrt{9/2} = \frac{3\sqrt{2}}{2}$$

distance of P from L